1 Conclusion

Summary of Contributions

This thesis has studied the interaction between randomized approximation and mixed-precision arithmetic in numerical linear algebra, using overdetermined least-squares problems as the model setting and regularized covariance transport as the main application-facing method.

Three interlocking contributions were developed:

1. **Perturbation analysis for regularized covariance transport.** We gave deterministic first-order bounds for the perturbation of the regularized transport map $T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2}$ under additive disturbances to the source and target covariance operators. The bounds make explicit how the amplification factor $K_\lambda(C_X, C_Y)$ depends on the regularized spectral gap, and they apply whether the perturbations arise from randomized low-rank compression, floating-point arithmetic, or both.

2. **Randomized low-rank approximation and mixed-precision error decomposition.** We decomposed the total covariance perturbation into a randomized truncation term governed by the spectral tail beyond the chosen rank, and a floating-point term governed by the unit roundoff, the accumulation length, and the norm of the sketching operator. This decomposition is the natural quantitative setting for the balancing-inexactness principle: mixed precision is safe only while $\|E_{\mathrm{fp}}\| \lesssim \|E_{\mathrm{rand}}\|$, and the regime boundary is computable from the rank, the oversampling, and the spectrum of the covariance operator.

3. **Numerical regime identification.** We designed experiments that vary the approximation rank, the regularization level λ, and the working precision in the sketch products, while tracking covariance error, transport-map error, and the induced dependence error on corrected future data. The experiments distinguish three regimes: an approximation-dominated regime in which lower precision is hidden beneath truncation error, a regularization-dominated regime in which transport instability is the bottleneck, and a precision-visible regime in which floating-point error overtakes the randomized approximation. The current implementation operates primarily in the first two regimes; the precision-visible crossover has not yet been reached experimentally.

Main Findings

On the least-squares side, the experiments confirm the expected picture. Residual preservation improves steadily with the sketch dimension, and the solution error is governed by the conditioning of the sketched matrix rather than by the residual quality alone. For Gaussian sketches with $n = 1024$ and $d = 20$ and moderate additive noise, the residual ratio falls below 1.1 only when $s \geq 200$, while the solution error decreases more rapidly with the sketch dimension. This confirms that residual and solution diagnostics probe different aspects of the perturbation structure.

On the covariance-transport side, the most important finding is the dominant role of regularization. For an exponential spectrum with $p = 30$ and condition number 100, the transport error at $\lambda = 10^{-2}$ is approximately 2 at rank 5 and decreases to approximately 1 at rank 25. At $\lambda = 10^{-6}$, the transport error remains above 80 even at rank 25. The covariance error, by contrast, is between 0.6 and 0.9 across all ranks and regularization levels. This confirms the theoretical prediction that the inverse-square-root amplification factor, scaling as $\lambda_{\min}(C_X + \lambda I)^{-3/2}$, is the decisive bottleneck.

The precision comparison shows that single and double precision produce nearly identical results in all tested configurations. The maximum relative difference in transport error is below 2% at rank 20 and $\lambda = 10^{-6}$. This is expected: the floating-point contribution is hidden beneath the much larger randomized truncation and regularization-amplified errors.

Limitations

The randomized covariance-transport implementation is not yet competitive with the deterministic full baseline on the synthetic benchmarks tested. The deterministic full covariance transport remains more accurate in correlation and future-sample error than the low-rank randomized variants. This is not a failure of the covariance-transport viewpoint itself. Rather, it identifies the approximation regime as the current bottleneck: the rank and regularization choices that make the low-rank surrogate accurate enough for transport-map perturbations to be small have not yet been reached.

A second limitation is that the precision-visible regime has not been observed experimentally. The experiments operate in regimes where truncation error and regularization amplification both dominate the arithmetic floor, so the balancing-inexactness prediction cannot be fully validated numerically.

The climate-model application chapter provides a workflow and validation design; full real-data benchmarking remains future work. The transition from synthetic covariance operators to real climate-data pipelines would require data access, preprocessing, and domain-specific diagnostics that are outside the scope of the present numerical work.

Future Work

Natural directions for later work include:

- Pushing the approximation rank and oversampling further, using power iteration or block Krylov methods in the range finder, to reach the regime where randomized truncation error is small enough that arithmetic precision becomes visible.
- A more detailed error analysis for particular sketch families in mixed precision, especially the SRHT, where the structured transform changes the accumulation model.
- More systematic comparisons of reduced-precision implementations on hardware that natively supports lower-precision arithmetic, rather than simulated rounding.
- A broader study of realistic climate-data pipelines, where the covariance-transport correction is embedded in a full post-processing workflow with domain-specific validation.
- Exploring the shared-basis variant further, including adaptive construction of the joint basis and its interaction with the regularization parameter.

The central conclusion of this thesis is that randomized numerical linear algebra and mixed precision are most informative when they are analyzed together. Randomized approximation and reduced precision must be tuned jointly: precision reduction becomes meaningful only after the randomized approximation and regularization parameters place the method in a credible accuracy regime.